

Merge 2 2D Arrays by Summing Values (Q2570)

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Approach 1

TC = $O(n+m) \log(n+m)$

- (S1) process the 1st 2D array, since all ids are unique, just build a hash table to store (id, value)
- (S2) now while processing 2nd array, if a id already exists in hash table then add curvalue to that id, otherwise just store (id, value) in hashtable
- (S3) fill ans array from hash table
- (S4) sort ans array

```

class Solution {
public:
    vector<vector<int>> mergeArrays(vector<vector<int>>& nums1,
vector<vector<int>>& nums2) {
        unordered_map<int,int> mp;

        vector<vector<int>> ans;
        int n=nums1.size();
        int m=nums2.size();

        for(int i=0;i<n;i++){
            mp[nums1[i][0]]=nums1[i][1];
        }

        for(int i=0;i<m;i++){
            if(mp.find(nums2[i][0])!=mp.end()){ //ye same id hamne nums1 me
dekh rkhi thi to add krna he bas
                mp[nums2[i][0]]+=nums2[i][1];
                continue;
            }

            mp[nums2[i][0]]=nums2[i][1]; //pehli baar dekh rhe to map me add
kar do

        }

        for(auto it:mp){
            int id=it.first;
            int val=it.second;

            ans.push_back({id,val});
        }
        sort(ans.begin(),ans.end());
        return ans;
    }
};

```

Approach 2 (Two Pointers)

TC = O(N)

Suppose			
Aadya		Arpit	
i → 1	2	j → 2	10
3	4	5	8
5	6	7	1

compare only current ids

(S1) $i < j : (1 < 2) \rightarrow$ so push (1, 2) to ans and move i to 3

(S2) now $j < i : (2 < 3) \rightarrow$ so push (2, 10) and j to 5

(S3) now $i < j : (3 < 5) \rightarrow$ so push (3, 4) and i to 5

(S4) both ($i == j == 5$) $\rightarrow$ so sum their values and push

```

class Solution {
public:
    vector<vector<int>> mergeArrays(vector<vector<int>>& nums1,
vector<vector<int>>& nums2) {
        int n=nums1.size();
        int m=nums2.size();

        int i=0;
        int j=0;
        vector<vector<int>> ans;
        while(i<n && j<m){
            if(nums1[i][0]<nums2[j][0]){
                ans.push_back({nums1[i][0],nums1[i][1]});
                i++;
            }

            else if(nums2[j][0]<nums1[i][0]){
                ans.push_back({nums2[j][0],nums2[j][1]});
                j++;
            }

            else{
                ans.push_back({nums1[i][0],nums1[i][1]+nums2[j][1]});
                i++;
                j++;
            }
        }

        while(i<n){
            ans.push_back({nums1[i][0],nums1[i][1]});
            i++;
        }

        while(j<m){
            ans.push_back({nums2[j][0],nums2[j][1]});
            j++;
        }

        return ans;
    }
};

```